



TRAINHUB



A connected gym experience for members and professionals



Teacher

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Academic Year

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The path from problem to working product

Six report chapters, presented as one continuous design and development process.

01 Introduction

Domain, initial scope and target users.

02 Competitor analysis

Existing ecosystems, fragmented alternatives and market gap.

03 User research

Questionnaire evidence, affinity mapping, personas and scenarios.

04 Application design

Navigation, paper prototype, Figma screens and design decisions.

05 Implementation

Architecture, offline behaviour, security and device features.

06 Conclusions

Project outcome and the most realistic future developments.

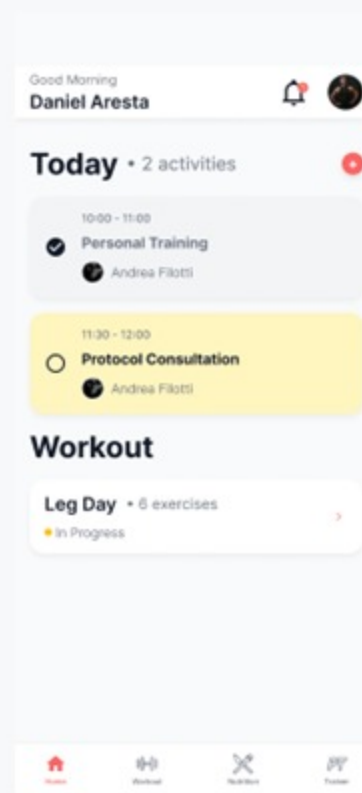
A single gym journey is split across unrelated tools

The user experiences one activity; the digital ecosystem treats it as several separate problems.

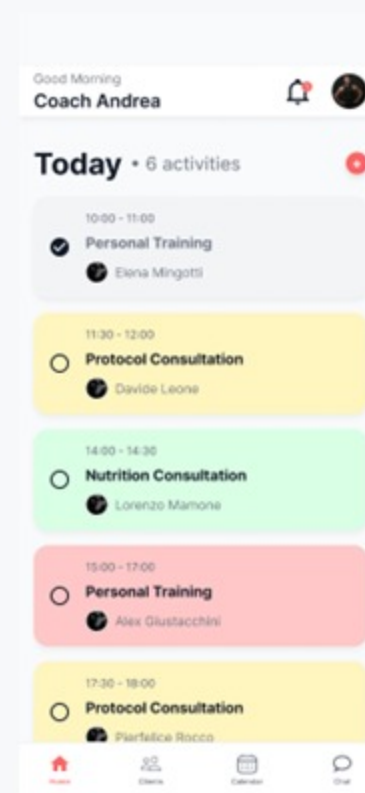
- Workout notes are often kept on paper, in phone notes or in a specialized tracking application.
- Nutrition plans frequently arrive as PDFs and are disconnected from appointments, progress and feedback.
- Members and professionals use generic messaging to exchange updates that quickly become difficult to retrieve.
- Facility access, bookings and subscription information usually belong to yet another closed gym system.

The consequence

More switching, duplicated information and less continuity between a workout and the professional response.



Member context



Professional context

TrainHub connects the core activities without becoming a gym ERP

The preliminary scope focused on actions that link members directly to their facility and professionals.

- 01 Workout** Members follow or create plans and record individual sets during a live training run.
- 02 Nutrition** Qualified professionals create plans; members consult the meals assigned to each weekday.
- 03 Appointments** Members request sessions and consultations while professionals manage availability and status.
- 04 Access** A rotating QR badge replaces the physical membership card and records validated check-ins.

GYM MEMBERS

Need fast input, clear daily information and direct access to their assigned professional.

GYM PROFESSIONALS

Need a client-centered view for planning, monitoring, communication and scheduling.

The analysis compared both integrated products and current substitutes

Applications were downloaded and tested, then compared by market dimensions and available features.

Direct competitors

Products tied to a gym or designed to manage the member–professional relationship.

- Virgin Active
- **EvolutionFit**
- Technogym
- Virtuagym

Indirect competitors

Specialized or generic tools that solve only one part of the same journey.

- Workout trackers
- Nutrition trackers
- Wearable and health platforms
- Notes, spreadsheets, PDFs and messaging

Two comparison levels

Business model, audience and platform reach were considered alongside workout, nutrition, booking, access, chat and staff tools.

Direct competitors demonstrate demand, but with different trade-offs

All cover workout and booking; nutrition, direct communication and staff tooling are less consistent.

Feature	Virgin	EvolutionFit	Technogym	Virtuagym
Workout tracking	YES	YES	YES	YES
Nutrition management	—	YES	—	YES
Booking	YES	YES	YES	YES
Facility access	YES	YES	YES	YES
Direct professional chat	—	YES	—	YES
Staff dashboard	—	YES	YES	YES

- **Virgin Active** offers a polished but closed chain-specific ecosystem.
- **EvolutionFit** is broad and gym-oriented, but its experience depends on the adopting facility.
- **Technogym** excels in training content, devices and wellness integration.
- **Virtuagym** provides the broadest suite, at the cost of greater product complexity.

Market validation The feature set is credible; the opportunity lies in a clearer and lighter integration.

The market gap is continuity, not feature count

TrainHub combines the essential flows while keeping the member experience mobile-first.

- Specialized applications are often excellent at one task, but require users to reconstruct the overall picture themselves.
- Large B2B platforms cover many business processes, yet can introduce complexity unrelated to the everyday gym journey.
- TrainHub places workout management at the center and connects nutrition, appointments, chat and facility access around it.
- The same product serves both roles, so a member action can immediately become useful professional context.

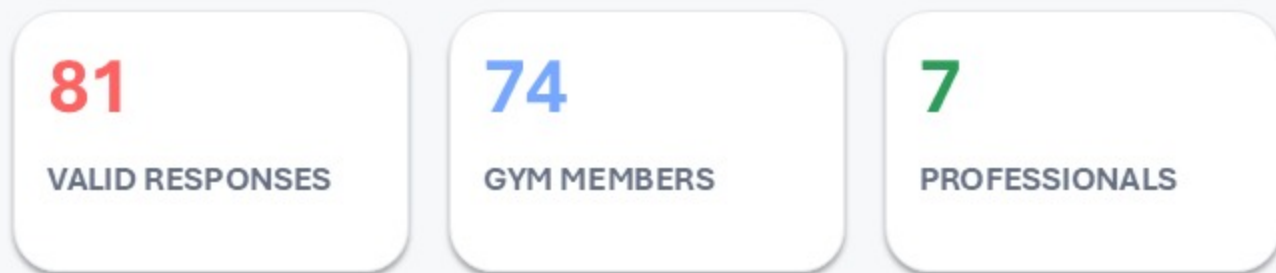
POSITIONING

**Integrated enough
to replace fragmentation.**

**Focused enough
to remain understandable.**

A branched questionnaire tested the initial assumptions

The questions changed according to role and combined quantitative answers with open-ended feedback.



The sample represents the two sides of the ecosystem, but professional percentages are interpreted cautiously because only seven professionals responded.

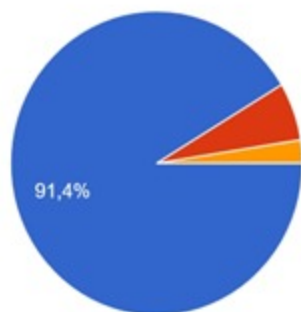
- Members described training frequency, tracking habits, application switching and their relationship with professionals.
- Professionals described client volume, monitoring tools, communication practices and recurring management difficulties.
- Open questions captured application names and problems that predefined answer lists might have missed.
- Early feedback was used to clarify ambiguous questions while data collection was still in progress.

The sample is young, active and predominantly member-side

Demographic results help define the main audience without excluding less digitally confident users.

Qual è il tuo ruolo principale nel mondo del fitness?

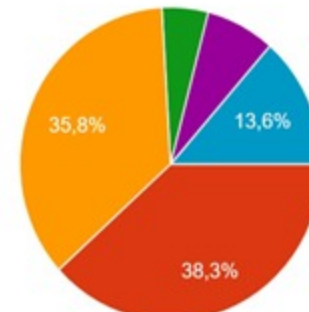
81 risposte



- Frequento (o ho frequentato) la palestra (Utente/Cliente)
- Sono un professionista (Personal Trainer/Nutrizionista/Dietista/Dietologo)
- Sono un professionista (Personal Trainer/Nutrizionista/Gestore)

Età

81 risposte



- Under 18
- 18-24
- 25-34
- 35-44
- 45-54
- 55+

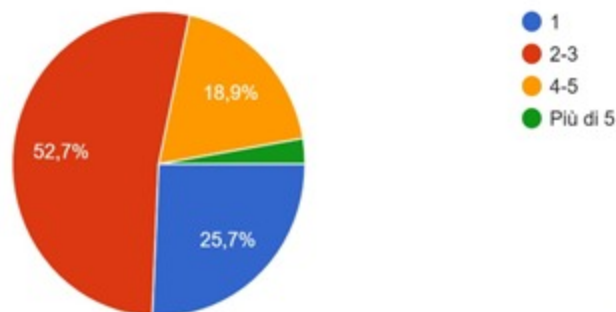
- Gym members account for 91% of responses, while professionals represent the operational minority expected in a physical facility.
- Most respondents fall within the 18–34 age range and are already accustomed to frequent smartphone use.
- The personas nevertheless include an older member to prevent the design from assuming uniform digital confidence.

Members train frequently, but tracking remains inconsistent

Activity level does not automatically translate into a structured digital routine.

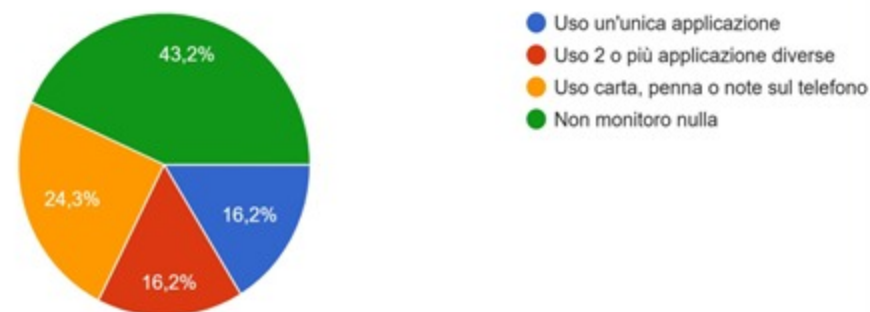
Quante volte a settimana ti alleni in media?

74 risposte



Attualmente, come monitori il tuo percorso (allenamento, dieta, prenotazioni)?

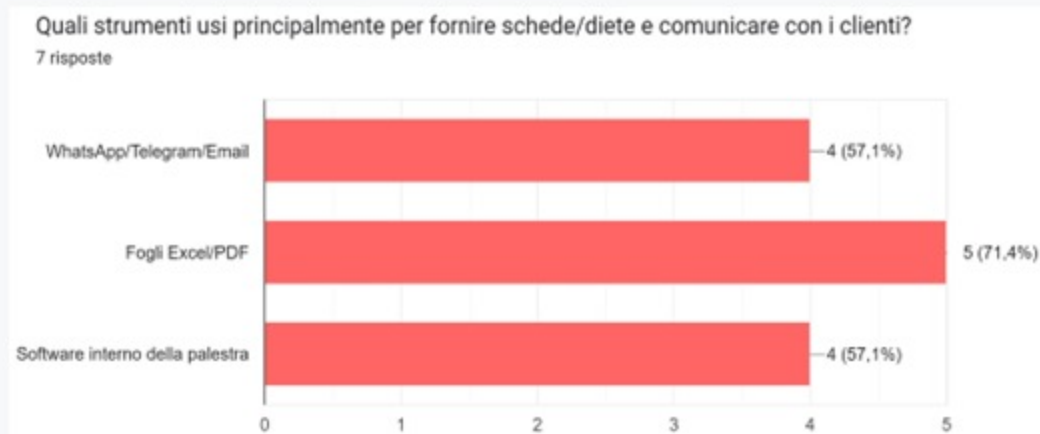
74 risposte



- Most respondents train between two and five times per week, so speed and repeatability are everyday requirements.
- Paper, phone notes and generic tools remain common; some users do not maintain a structured record at all.
- Switching among multiple applications creates friction precisely during the moments that should require the least attention.
- A direct digital connection with the trusted professional was considered especially valuable for plans and feedback.

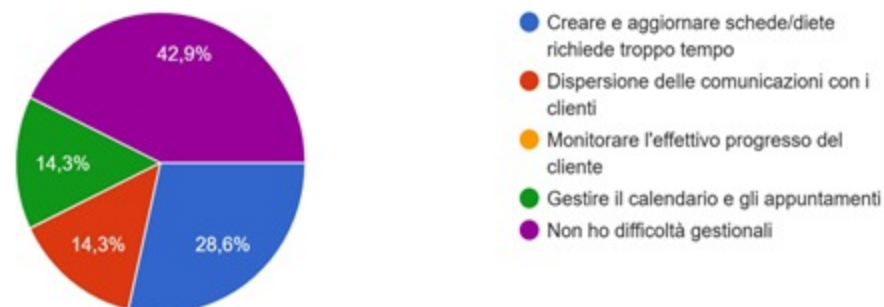
Professionals still assemble client management from generic tools

The results are directional, but they reveal the same fragmentation from the operational side.



Qual è la tua più grande difficoltà gestionale ad oggi?

7 risposte



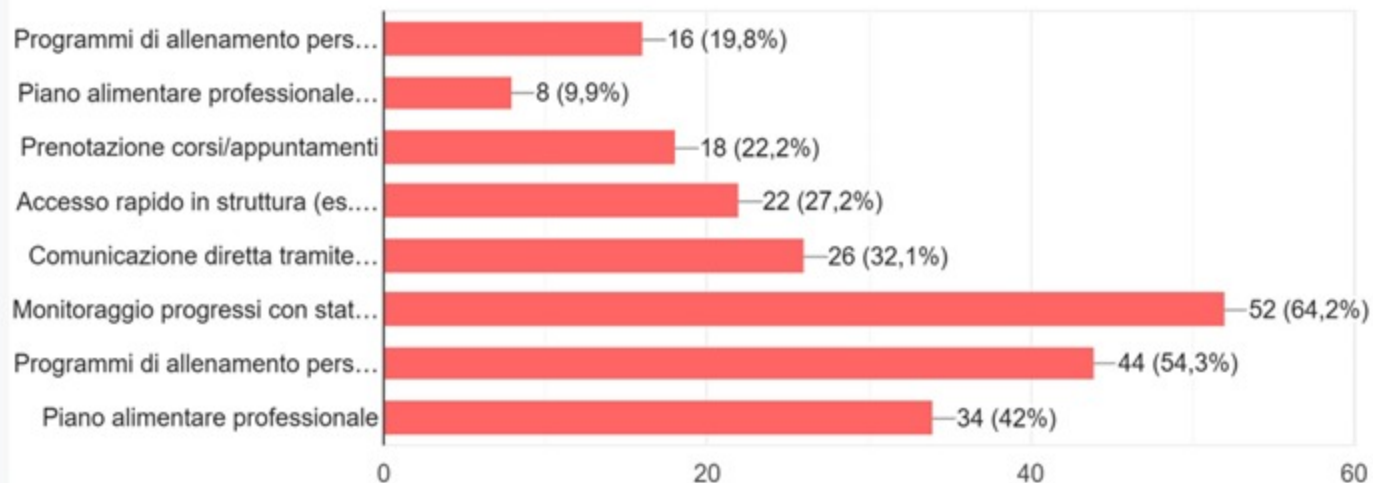
- Spreadsheets and PDFs are used by 71% of respondents; messaging and internal gym software are each used by 57%.
- The most frequent concrete difficulty is the time required to create and update workout or nutrition plans.
- Communication dispersion and appointment management also appear, although each percentage represents very few people.
- TrainHub therefore consolidates the workflow without claiming that the small professional sample represents the entire market.

The expected value is integration around three core capabilities

Respondents confirmed the central scope while also suggesting secondary services.

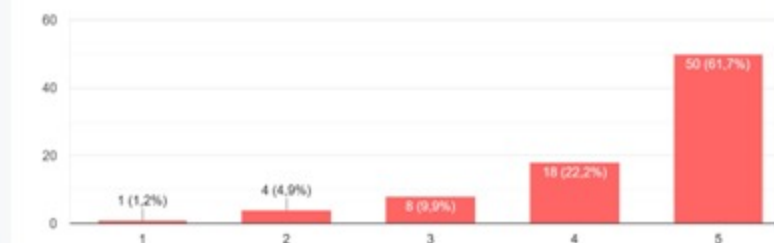
Quali tra queste funzionalità ritieni fondamentali?

81 risposte



Quanto troveresti utile un'unica app che integri tutti questi servizi?

81 risposte



74%

personalized workout tracking

64%

progress monitoring and statistics

52%

professional diet management

Affinity mapping turned scattered answers into design direction

Notes were clustered, labelled and linked before personas and scenarios were finalized.

01 Digital fragmentation

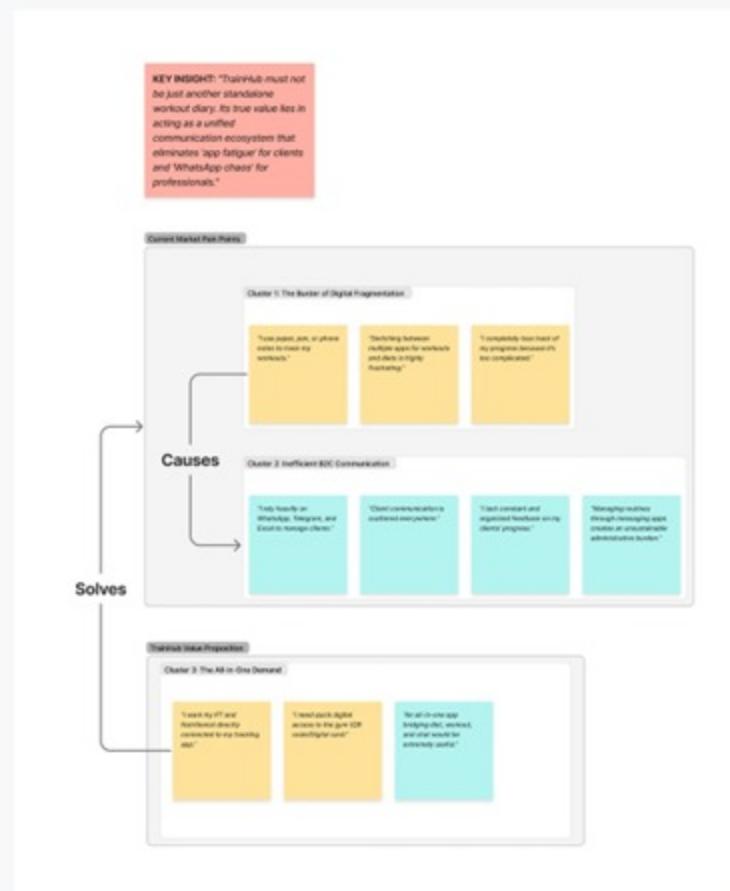
Users reconstruct one journey from notes, specialized apps and files.

02 Inefficient communication

Important plans and feedback become difficult to locate in generic chats.

03 All-in-one demand

The desired solution is not another tracker, but a shared operational context.



Personas represent different abilities, contexts and responsibilities

They provide concrete reference points for interaction choices rather than abstract role labels.



Marco Rovella

Age 26
 Profession Software Engineer
 Family Single
 City Milan
 Hobby Photography, hiking
 Analytical Goal-oriented Tech-savvy

Description
 Marco goes to the gym regularly but struggles to keep track of his routines and diet. He currently uses his phone's notes app because generic fitness apps feel too disconnected from his actual personal trainer and take too long to update during intense workouts.

Motivation
 Wants a streamlined, fast way to view his custom workout plans, track his progress, and communicate with his trainer seamlessly in one single place.

Frustrations

- Discontinuity across different tracking tools
- Slow data input in standard apps
- Lack of a unified digital experience in the gym

Main use
 Tracking customized workouts, checking diet plans, booking gym classes or PT sessions, and accessing the facility.

Frequency of use
 4-5 times a week.

Quote
 "I just want to focus on my workout without juggling three different apps and my phone's notes."

Marco

YOUNG MEMBER

Wants immediate set logging during a time-constrained workout.



Roberto Ferrari

Age 52
 Profession Bank Branch Manager
 Family Married, two teenage children
 City Brescia
 Hobby Cycling on weekends, reading
 Pragmatic Health-conscious Reliable

Description
 Roberto joined the gym recently, partly on his doctor's advice to manage stress and improve his posture. He relies heavily on his personal trainer because he is afraid of doing exercises incorrectly and getting injured.

Motivation
 Having a direct line of communication with his trainer gives him the peace of mind that he is being supervised, even when the trainer isn't physically next to him.

Frustrations

- Dealing with confusing booking systems
- Feeling overwhelmed by overly complex fitness apps that track too many unnecessary metrics

Main use
 Booking stretching/restoral classes, checking simple video instructions for his exercises, reading the diet plan, and chatting with his trainer for reassurance.

Frequency of use
 2-3 times a week.

Quote
 "I want to stay active and healthy, but I need a professional to guide me safely without complicating my daily routine."

Roberto

SENIOR MEMBER

Needs readable guidance and reassurance before training independently.



Elena Bonetti

Age 34
 Profession Personal Trainer
 Family Married, one child
 City Rome
 Hobby Crossfit, listening to podcasts
 Energetic Hands-on Motivating

Description
 Elena is constantly on the gym floor moving between clients. She spends her evenings writing training programs. She relies on messy communication channels (like WhatsApp) to get feedback on how heavy the weights felt for her clients.

Motivation
 Needs a fast, centralized hub to assign workout plans and see immediately if her clients are hitting their target weights and reps.

Frustrations

- Clients losing their paper cards
- Not knowing if a client skipped a workout until the end of the week
- Messy WhatsApp updates

Main use
 Creating workout plans, tracking lifting progress, managing floor appointments

Frequency of use
 Daily.

Quote
 "I need to see my clients' progress on the gym floor without scrolling through endless chat logs."

Elena

PERSONAL TRAINER

Reviews several clients and adjusts plans during short breaks.



Matteo Costa

Age 41
 Profession Sports Nutritionist
 Family Cohabitating, no children
 City Turin
 Hobby Cooking, hiking
 Analytical Organized Empathetic

Description
 Matteo works in an office inside the gym facility. He creates highly customized meal plans based on client goals (bodybuilding, weight loss). He struggles to get clients to report their weight and adherence accurately.

Motivation
 Wants a clean digital space to upload diet plans, track clients' weekly body metrics, and adjust macros quickly without sending out new PDFs every time.

Frustrations

- Clients forgetting to send their weekly check-ins
- Having to completely rewrite a PDF diet plan just to change a single snack

Main use
 Creating and updating diet plans, monitoring client weight/measurements, answering nutrition-related questions

Frequency of use
 Daily.

Quote
 "Customizing diets for 40 different people requires precision and an organized system, not messy PDF files."

Matteo

NUTRITIONIST

Checks measurements and updates meal plans from a larger screen.

Scenarios translated personas into interaction requirements

Each scenario combines a real context, a user goal and an observable design response.

CONTEXT	GOAL	DESIGN RESPONSE
Marco · gym floor	Log prescribed weight without interrupting concentration.	Prefilled values, fast set controls and persistent live run.
Roberto · at home	Understand tomorrow's workout and ask for reassurance.	Readable session details and direct access to the assigned trainer.
Elena · short break	Review progress and adjust a routine in a few minutes.	Client-centered progress and versioned plan creation.
Matteo · office	Use measurements to update nutrition before the weekend.	Nutrition editor organized by day types, meals and macros.

Research priorities became explicit interface principles

The design phase established hierarchy and navigation before visual details were refined.

01 Speed during training

Frequent workout actions remain visible, large and close to the current exercise.

02 Client-centered professional work

Plans, progress and communication branch from the selected client rather than separate silos.

03 One product, distinct roles

Shared services stay consistent while navigation and permissions reflect different responsibilities.

04 Context before confirmation

Booking and plan editing preserve the selected day, professional or client until the final action.

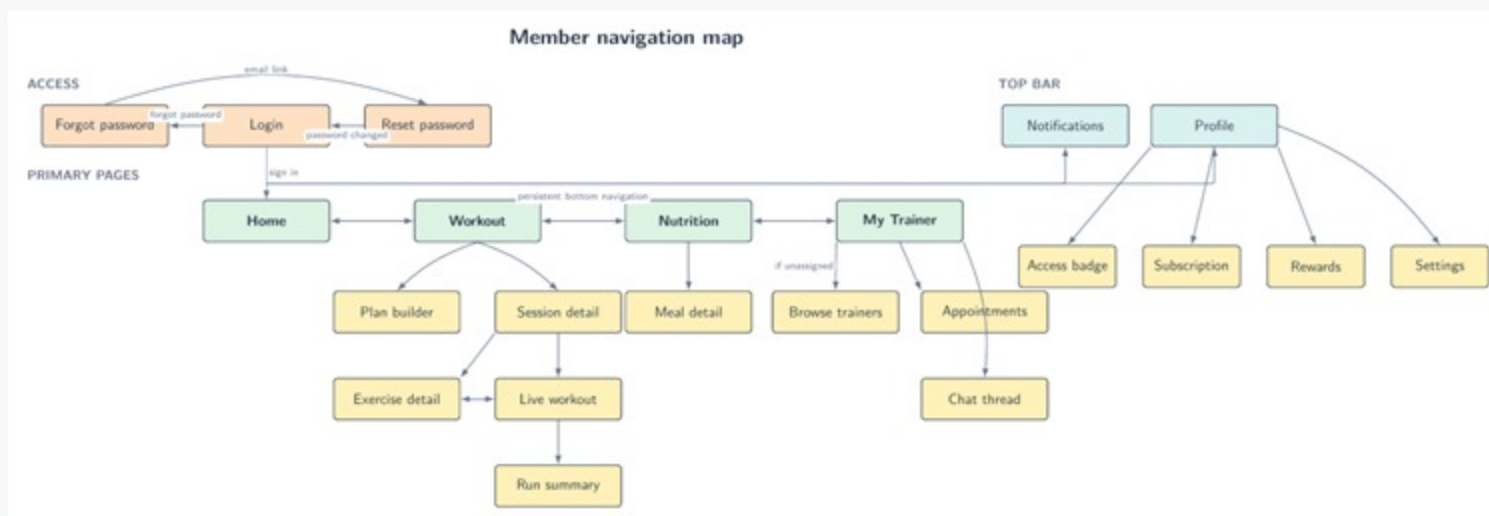
Design sequence

Navigation map → paper prototype → Figma prototype → implementation feedback

Member navigation prioritizes daily activities

The bottom bar exposes four frequent destinations; the header contains global account actions.

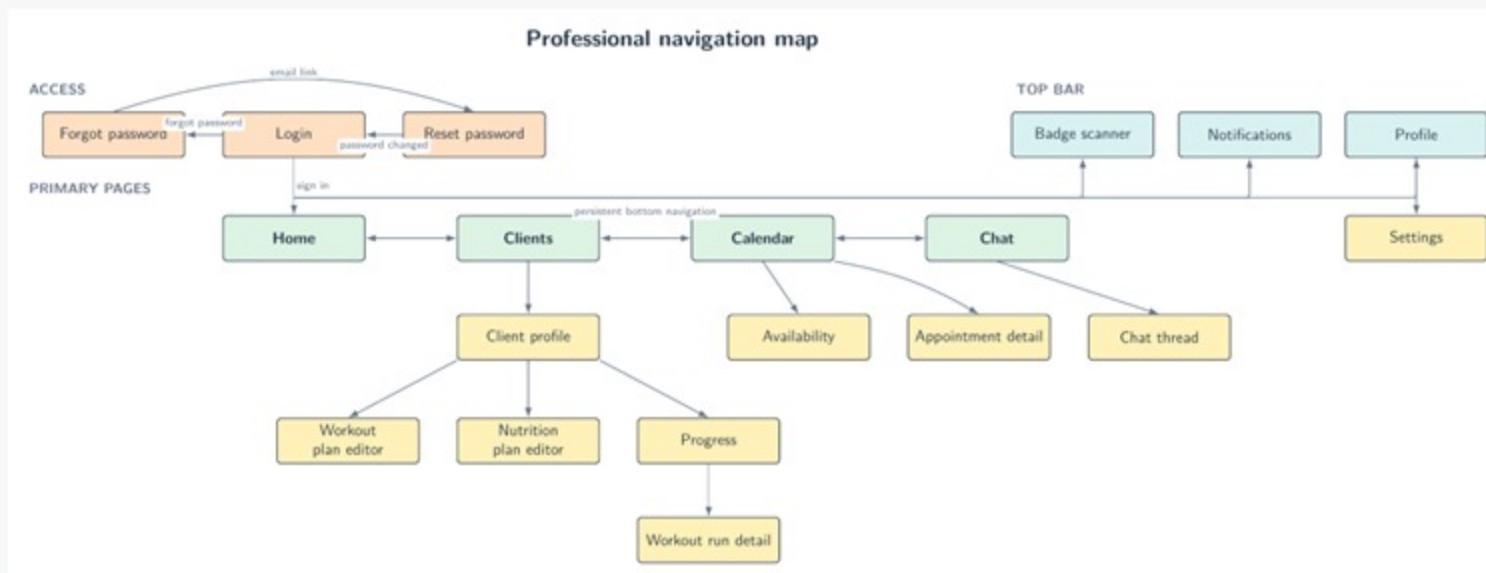
- **Home** summarizes the current day, next appointments and workout context.
- **Workout** leads from the active plan to sessions, exercises and live runs.
- **Nutrition** presents the meals assigned to the selected weekday.
- **My Trainer** connects professional discovery, appointments and chat.
- **Profile** groups badge, subscription, rewards and settings outside the daily path.



Professional navigation is organized around clients and time

Client management and calendar activities remain separate but share communication and notifications.

- **Home** focuses on the daily agenda and actions requiring attention.
- **Clients** opens each person's plans, progress, chat and membership state.
- **Calendar** combines appointments, status updates and recurring availability.
- **Chat** provides direct access to ongoing member conversations.
- **Header actions** expose profile, notifications and the professional QR scanner.



Paper prototypes checked flow depth before visual styling

Small sketches made structural mistakes cheap to identify and correct.



Member workout flow



Professional client flow



Professional calendar

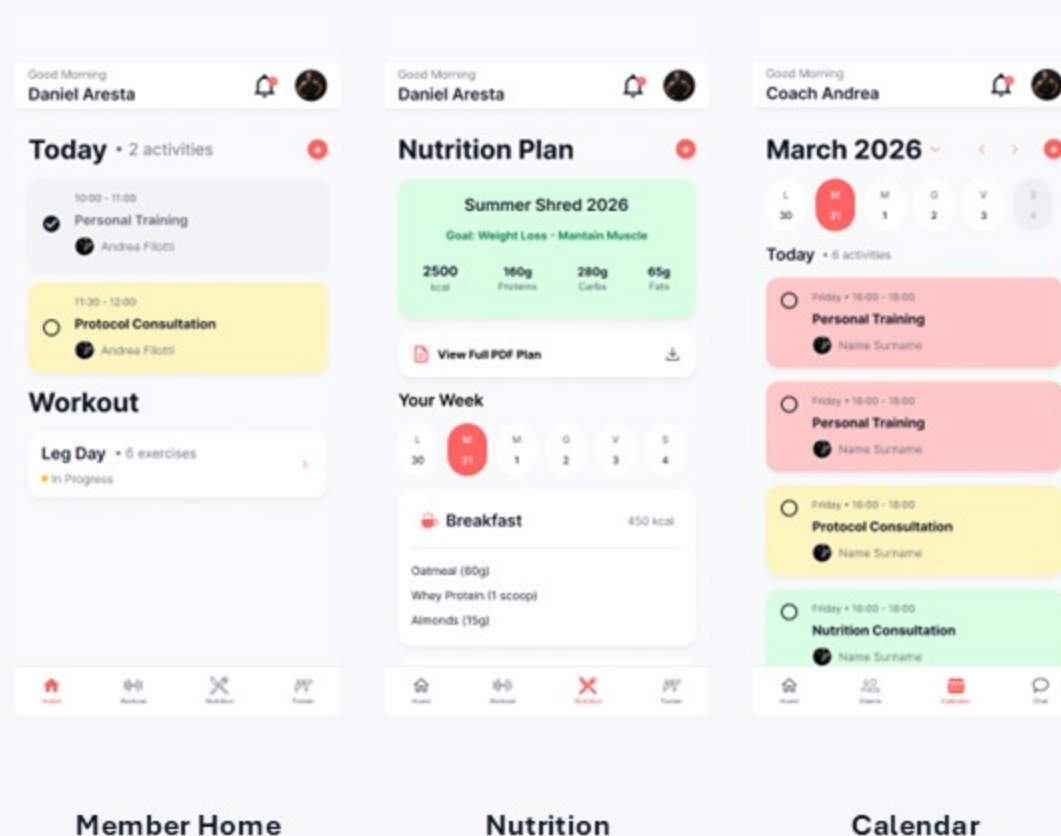
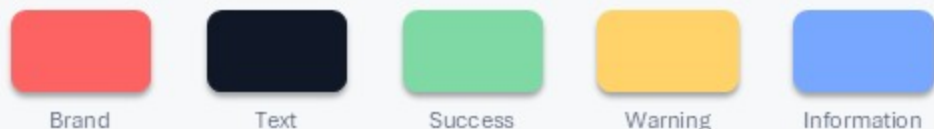
- The member sketches connected plan exploration, exercise details and set recording without adding decorative detail.
- The professional sketches established the selected client as the entry point for planning and progress.
- Repeated headers, cards, status labels, calendars and primary actions became candidates for reusable components.

The visual system supports fast reading and consistent states

Figma tokens were carried into the Material UI theme instead of being repeated screen by screen.

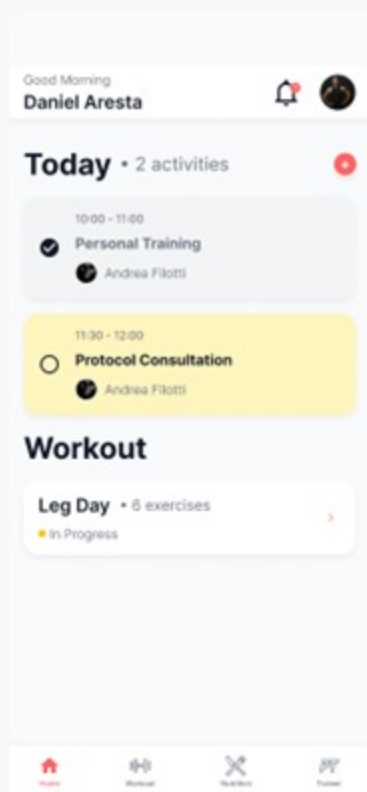
Visual rules

- Inter establishes a clear hierarchy between page titles, section headings, body text and secondary labels.
- Rounded cards and pill actions create a soft identity without reducing contrast or touch-target size.
- Semantic tokens distinguish success, warning, danger, appointments and membership states consistently.
- Color is never the only signal: labels, icons and numeric values repeat the same meaning.



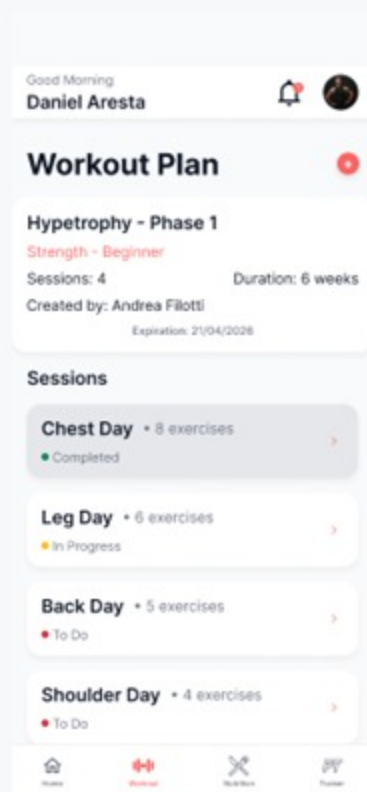
The member prototype presents overview first, detail on demand

Five representative screens show how the daily experience remains coherent across domains.



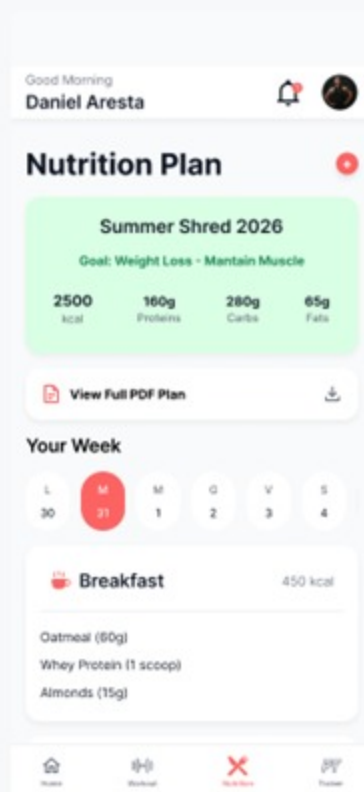
Today

Immediate context



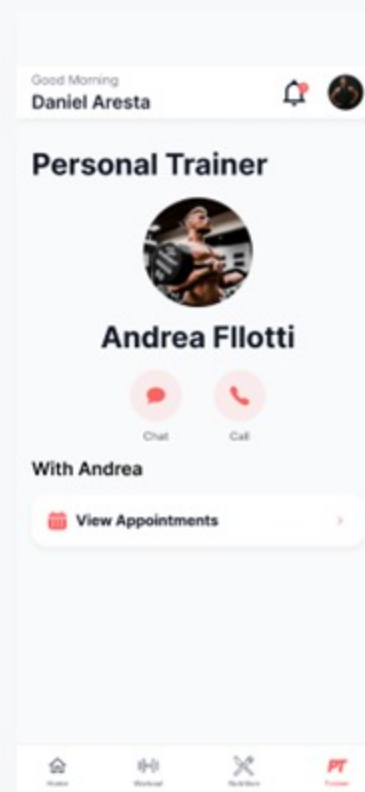
Workout

Plan and sessions



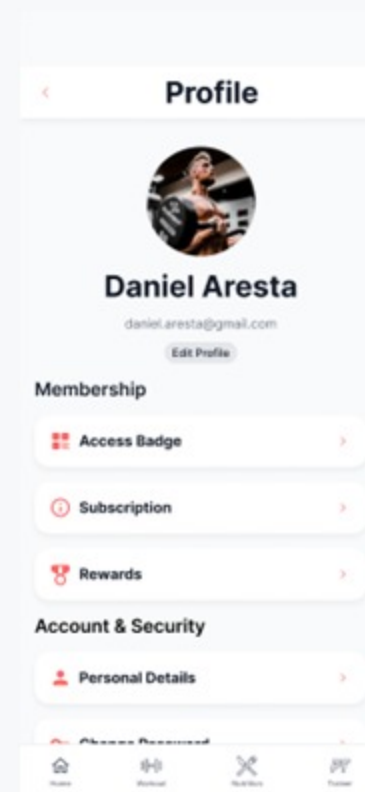
Nutrition

Meals by weekday



My Trainer

Chat and booking

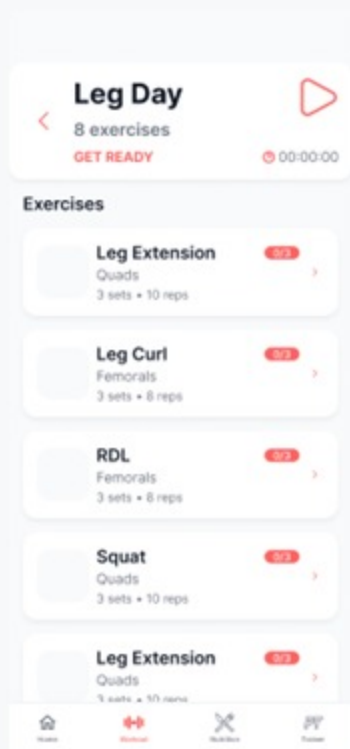


Profile

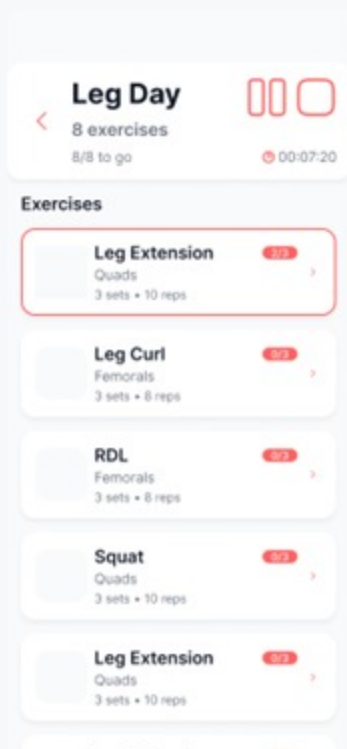
Membership services

Live workout is the central interaction flow

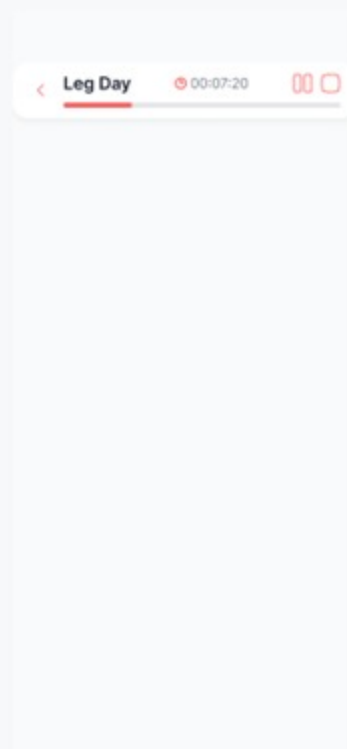
The interface keeps the current run available while progress is derived from completed sets.



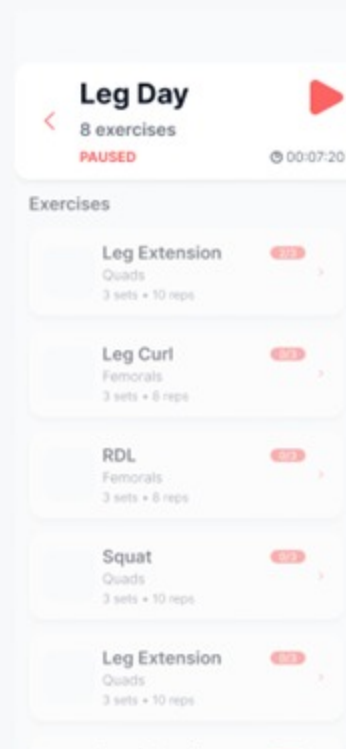
1. Review the prescription



2. Start a persistent run



3. Log a completed set



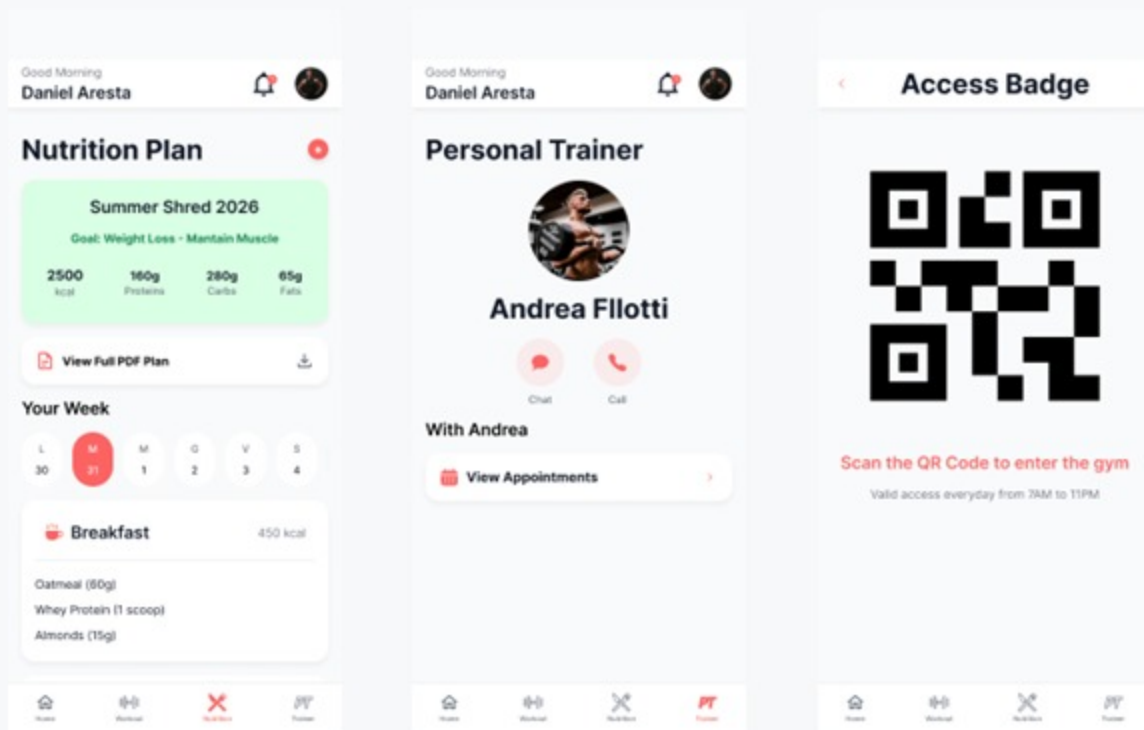
4. Pause or resume safely

Run summary

- Duration excludes pauses.
- Completion uses valid logged sets.
- Outcome can be completed, partial or abandoned.
- Points respect membership status.
- An optional note reaches the professional.

Secondary member services remain connected to the same context

Nutrition, booking and access support the workout journey without competing with it.



Nutrition plan

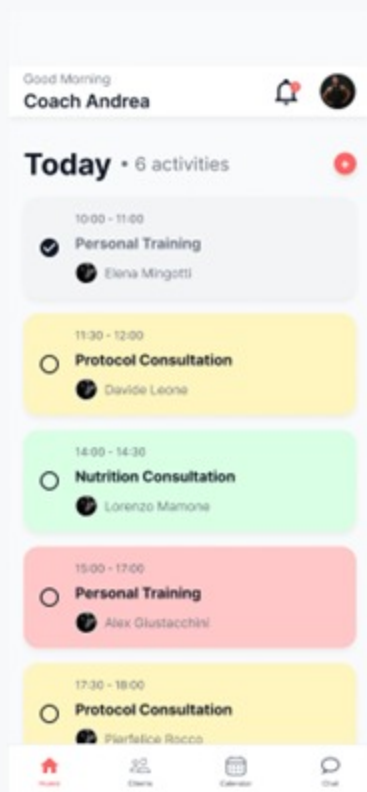
Assigned trainer

Access badge

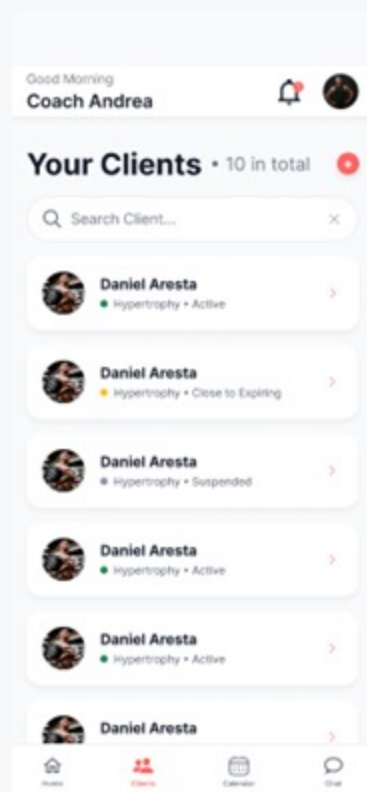
- Selecting a weekday changes the nutrition day type and the meals shown for that date.
- Appointments are created in a bottom sheet so the chosen trainer and day remain visible.
- The rotating badge uses a short-lived token and provides a manual code as an accessible fallback.
- Subscription and rewards are available from Profile, outside the persistent bottom navigation.

The professional interface moves from agenda to individual evidence

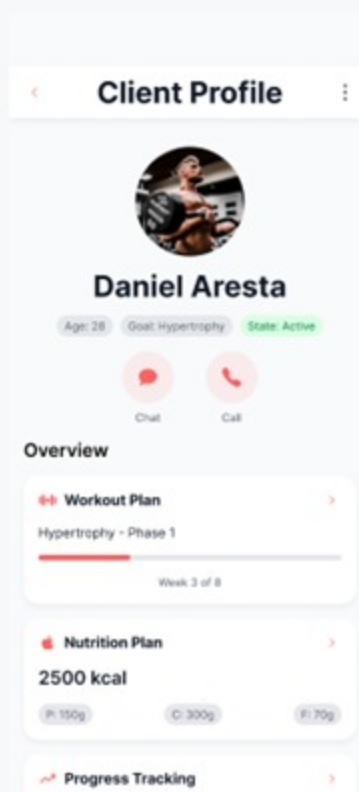
The selected client remains identifiable while the professional changes plans or reviews progress.



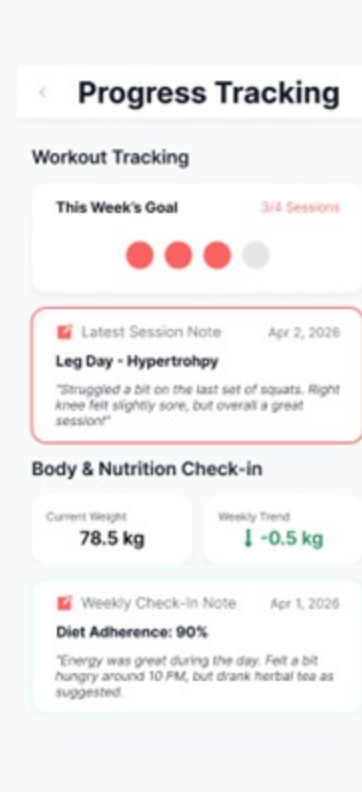
Agenda



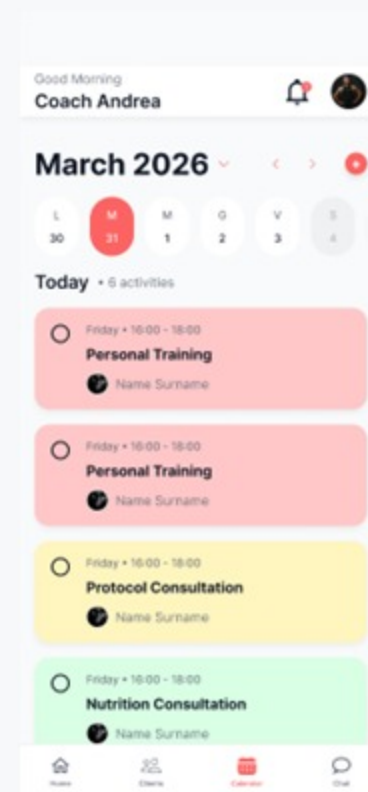
Client list



Client hub



Progress



Calendar

Implementation refined several assumptions from the wireframes

The purpose of the original flows remained stable while real data introduced new constraints.

One professional role Specialty determines which planning actions are relevant.

Profile in the header Four primary destinations keep the bottom navigation clear.

Contextual booking A bottom sheet preserves the selected person and date.

Weekly workout runs The same prescribed session can be trained more than once.

Versioned plans Editing does not rewrite the meaning of completed workouts.

Draft-then-commit editors A complete plan is written only after final review.

Nutrition day types Weekday selection changes the meals displayed.

Persistent notifications Messages and domain updates share one center.

Membership enforcement Status now controls booking, access and rewards.

Exercise media deferred A licensed and coherent library remains future work.

The stack separates presentation, remote state and trusted rules

Each technology is used for a specific responsibility rather than as an interchangeable layer.

- | | | |
|----|----------------------------|--|
| 01 | React + Material UI | Responsive screens, reusable components and theme-based styling. |
| 02 | React Router | Public routes and protected member/professional application trees. |
| 03 | TanStack Query | Remote-state cache, invalidation, mutation defaults and persistence. |
| 04 | Supabase | Authentication, PostgreSQL, Realtime and Edge Functions. |
| 05 | Workbox + Web APIs | Installability, offline shell, push, camera and QR handling. |

Layering principle

**Screens describe interaction.
Data functions describe
requests.
The database protects stored
meaning.**

This avoids repeating filters, date conversion and authorization assumptions inside every page.

Application structure keeps domain logic reusable

Feature screens depend on shared data functions, query keys and a role-aware application shell.

Source organization

src/features	Domain screens for workout, nutrition, chat, calendar, profiles and progress.
src/components	Reusable navigation, feedback, status and card components.
src/data	Plain asynchronous functions for reads and secure operations.
src/lib	Supabase configuration, query keys, cache persistence and utilities.
src/routes	Public routes plus protected member and professional trees.

Remote-state strategy

- Shared query keys identify both the domain and the relevant user or record.
- Mutations invalidate every view affected by the same operation, including the other role's view.
- Offline mutation defaults are registered before cache restoration so queued operations can resume after restart.
- Ordered scopes preserve sequence for dependent writes such as consecutive workout sets.

The relational model preserves history and ownership

A plan describes the prescription; a run records what actually happened during one attempt.

Data model

- Workout and nutrition plans create replacement versions instead of overwriting records referenced by past activity.
- Workout runs store timestamps, pauses, logged sets, percentage, outcome and optional professional feedback.
- Communication, appointments, facility access and body measurements remain linked to authenticated profiles.

Security model

- Row Level Security limits members to their own data and professionals to assigned clients.
- Restricted grants prevent direct writes to tables that require coordinated validation.
- Security-definer database functions execute multi-table operations atomically and re-check the authenticated actor.
- Client-generated identifiers and version checks make retries safe without silently overwriting newer work.

The browser can hide an action; only the backend can make the rule unavoidable.

Offline support is selective and explicit

TrainHub remains useful without presenting stale or time-sensitive actions as reliable.

What remains available

- The service worker precaches the application shell, required assets and local fonts for installed startup.
- Previously loaded Supabase responses remain available from a time-limited IndexedDB query cache.
- Recoverable mutations retain their original identifier and timestamp and resume when connectivity returns.
- A cache migration removes queued payloads that no longer match the current database contract.

What requires a connection

- Authentication requires a current server session and cannot be simulated locally.
- Badge redemption must verify expiry, previous use, assignment and subscription at that moment.
- Push activation requires browser permission, a live push service and server registration.
- Offline banners and action-level messages state the limitation instead of silently failing.

Realtime and device features improve immediacy with graceful fallback

Permission-dependent capabilities never block workout, nutrition, booking or chat as a whole.

01

Chat and Realtime

Inserted messages and read receipts update the query cache immediately. Periodic refresh remains available when a mobile connection silently drops the socket.

Fallback

Refresh

02

Persistent and Web Push

Notifications are stored before delivery, so the in-app center remains complete even when push is unavailable. Invalid browser endpoints are removed.

Fallback

In-app history

03

QR badge and camera

The member receives a short-lived random token; the professional scans or enters it and the database validates every access condition.

Fallback

Manual code

The workout flow validates the cooperation of every layer

A complete scenario crosses routing, cache, secure operations, derived results and both role interfaces.

- 1** Load the active plan and current-week runs through the persisted query cache.
- 2** Create one secure run after checking ownership and preventing accidental duplicates.
- 3** Record ordered sets against both the run and the prescribed exercise.
- 4** Derive elapsed time from timestamps and accumulated pause time across screens.
- 5** Close the run and calculate percentage, outcome and eligible reward points in the database.
- 6** Refresh the member summary and the professional progress view from the returned result.

Verification

Lint and production build · focused self-checks · database inspection and authenticated probes · complete manual scenarios · installed PWA, camera and push on real devices

The result is a connected working relationship, not a collection of screens

Member autonomy and professional context now reinforce each other inside the same product.

Project outcome

- Members can move from facility access to planning, training, nutrition, appointments and communication without changing service.
- Professionals can create plans, review performed work, organize appointments, manage memberships and validate access.
- Updates retain their context: a completed workout becomes progress evidence and a new plan becomes an actionable member task.

Future developments

- Introduce a licensed and visually coherent library of exercise images or short videos.
- Add richer archives, reports and comparisons for long-term progress analysis.
- Connect subscription billing, opening hours and professional availability to real gym systems.
- Personalize notification frequency and content according to specialty, preferences and member goals.



THANK YOU

Questions and discussion

Davide Leone · Andrea Bariselli